25900240 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBIF201 - LINUX AND C MASTERY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Explain about the keywords in C. [U] / [CO1]
2. Discuss the role of the kernel in process management. [U] / [CO3]
3. Write the basic syntax for a `while` loop. [K] / [CO8]
4. Illustrate the use of `break` in a `for` loop. [A] / [CO2]
5. Describe the use of the *chmod* command in Linux. [U] / [CO4]
6. Explain the difference between scanf() and printf() functions. [U] / [CO1]
7. Create a pipeline that counts the number of lines in a file which contain a specific word. [A] / [CO5]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8. What do exit status codes represent in shell scripting? Give examples. [K] / [CO8]

9. Explain 1-D array with suitable example. [A] / [CO2]

10. Describe how *typedef* improves readability in user-defined data types. [U] / [CO1]

11. Write the command to check the connectivity to google.com. [A] / [CO6]

12. Explain the importance and method of making a shell script executable. [U] / [CO7]

13. Explain the purpose of conditional statements in shell scripting with an example. [K] / [CO8]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

14. Apply an if-else statement to check whether a number is positive or negative. [A] / [CO2]

15. a) Describe two ways to debug a shell script effectively. [K] / [CO8]
    b) Explain the role of `set -x` and `set +x` in debugging.

16. Explain why the documentation section is important in large C programs with an example. [U] / [CO1]

17. Explain how to change file permissions using chmod with an example. [A] / [CO6]

[10x2 = 20]